

EXP.NO: 7 write a servlet program to store a user's preferences(like theme or language)using cookies. Retrieve and display these preferences on subsequent visits

Aim

To develop a Java Servlet program that stores user preferences (such as theme and language) using cookies and retrieves them on subsequent visits to display the saved preferences.

Procedure

- 1.Create a Dynamic Web Project in your IDE (e.g., Eclipse).
- 2.Add Servlet API library (if not already configured).
- 3.Create a Servlet class (e.g., PreferenceServlet.java).
- 4.Design an HTML form to collect user preferences (theme, language).
- 5.Store preferences in cookies when the user submits the form.
- 6.Retrieve cookies when the user revisits the page.
- 7.Display stored preferences in the browser.
- 8.Deploy the project on a server like Apache Tomcat.
- 9.Run the application and test cookie persistence.

Code:

```
package com.demo;
```

```
import jakarta.servlet.ServletException;
```

```
import jakarta.servlet.annotation.WebServlet;
```

```
import jakarta.servlet.http.*;
```

```
import java.io.IOException;
```

```
@WebServlet("/CookieServlet")
```

```
public class CookieServlet extends HttpServlet {
```

```
protected void doPost(HttpServletRequest request, HttpServletResponse response)
throws ServletException, IOException {

String name = request.getParameter("username");

Cookie cookie = new Cookie("user", name);

cookie.setMaxAge(60*60); // cookie valid for 1 hour

response.addCookie(cookie);

response.sendRedirect("showCookie");

}

}
```

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Cookie Login</title>
```

```
<style>
```

```
body{
```

```
font-family: Arial;
```

```
background: linear-gradient(135deg,#667eea,#764ba2);
```

```
height:100vh;
```

```
display:flex;
```

```
justify-content:center;
```

```
align-items:center;
```

```
}
```

```
.container{  
background:white;  
padding:40px;  
border-radius:12px;  
width:350px;  
box-shadow:0 10px 25px rgba(0,0,0,0.2);  
text-align:center;  
}
```

```
h2{  
margin-bottom:20px;  
color:#333;  
}
```

```
input{  
width:100%;  
padding:10px;  
margin:10px 0;  
border-radius:6px;  
border:1px solid #ccc;  
}
```

```
button{  
width:100%;  
padding:12px;
```

```
background:#667eea;

color:white;

border:none;

border-radius:6px;

font-size:16px;

cursor:pointer;

}

button:hover{

background:#4c5bd4;

}

</style>

</head>

<body>

<div class="container">

<h2>Login Page</h2>

<form action="CookieServlet" method="post">

<input type="text" name="username" placeholder="Enter your name" required>

<button type="submit">Login</button>

</form>

</div>

</body>

</html>

package com.demo;
```

```
import jakarta.servlet.ServletException;

import jakarta.servlet.annotation.WebServlet;

import jakarta.servlet.http.*;

import java.io.*;

@WebServlet("/showCookie")

public class ShowCookieServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter out = response.getWriter();

        Cookie cookies[] = request.getCookies();

        String name = "Guest";

        if(cookies != null){
            for(Cookie c : cookies){
                if(c.getName().equals("user")){
                    name = c.getValue();
                }
            }
        }

        out.println("<html>");
```

```
out.println("<head>");

out.println("<title>Welcome</title>");


out.println("<style>");

out.println("body{font-family:Arial;background:#f2f2f2;text-align:center;padding-top:120px;}");

out.println(".card{background:white;padding:40px;border-radius:12px;width:400px;margin:auto;box-
shadow:0 10px 25px rgba(0,0,0,0.2);}");

out.println("a{display:inline-block;margin-top:20px;text-
decoration:none;background:#667eea;color:white;padding:10px 20px;border-radius:6px;}");

out.println("</style>");

out.println("</head>");

out.println("<body>");

out.println("<div class='card'>");

out.println("<h2>Welcome " + name + "</h2>");

out.println("<p>You are successfully logged in using Cookies.</p>");

out.println("<a href='index.html'>Logout</a>");

out.println("</div>");

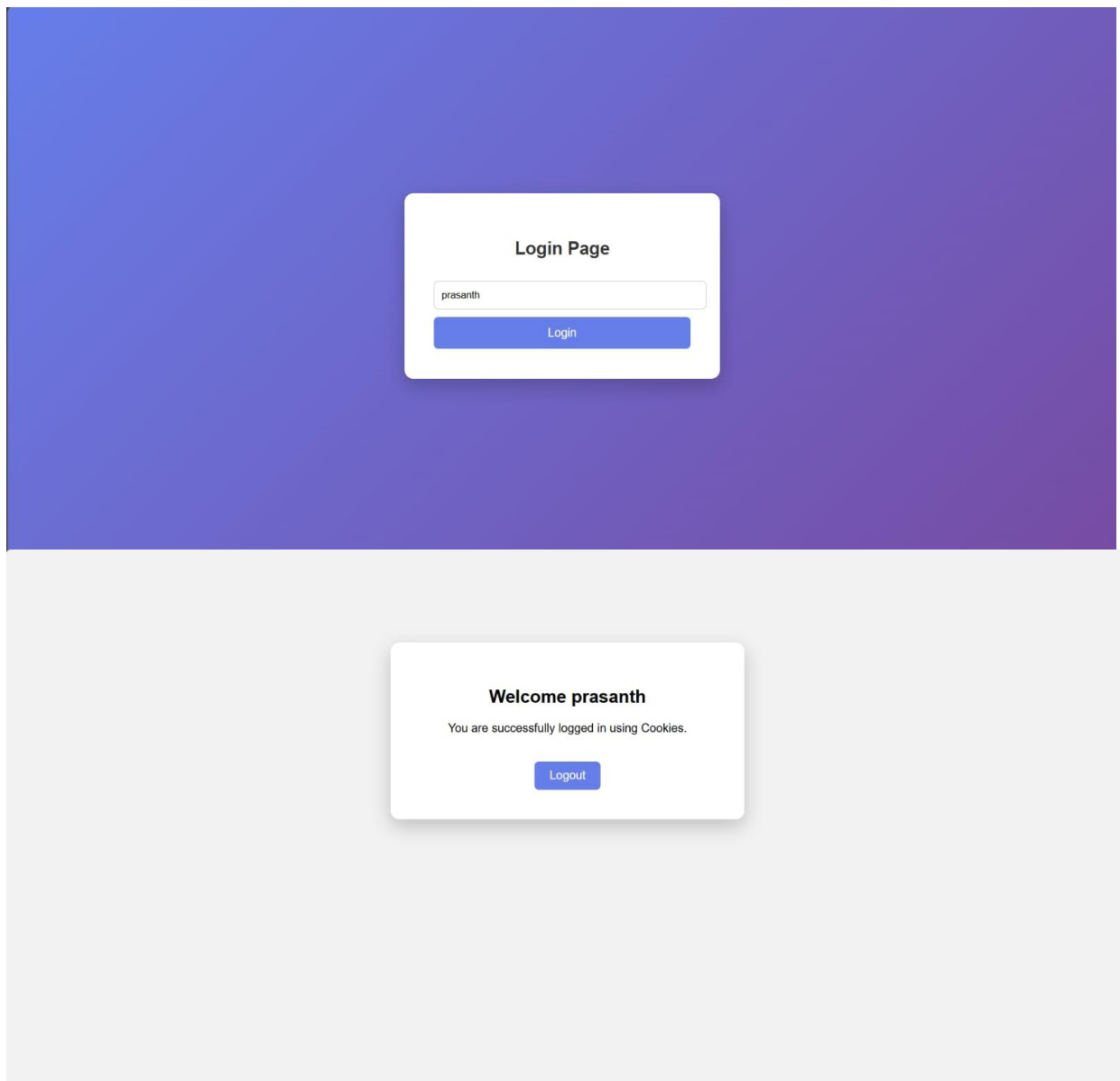
out.println("</body>");

out.println("</html>");

}

}
```

Output:



Result

The servlet program successfully stores user preferences (theme and language) using cookies. When the user revisits the application, the stored preferences are retrieved from the cookies and displayed correctly in the browser.